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A multicomponent, dual-polarity palaeomagnetic regional overprint from the Moine of northwest Scotland *

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Palaeomagnetic remanence may be reset by orogeny in a manner analogous to isotopic systems. The heavy mineral seams and associated psammite in the deformed and metamorphosed Lower Morar Division of the Moine Assemblage of northwest Scotland have a palaeomagnetic signature that is characteristic over a wide area. The various components of the magnetization are resolved by detailed thermal demagnetization. These components are post-tectonic, as they are not affected by any structural fabric. They may be classified according to blocking temperature and, when statistically distinguishable, show a systematic change of direction from high to lower temperature possibly associated with apparent polar wander. Comparison to the European palaeomagnetic record suggests a late Silurian to early Devonian age of magnetization for the localities south of the Strath Conon Fault. The distribution of remanance directions and blocking temperatures is not consistent with the pattern of potassium-argon ages in this part of the British Caledonides. If the magnetization is purely of thermal origin, the palaeomagnetic record indicates higher temperatures at later time than does the potassium-argon record. This remagnetization event is related to the closing stages of the Caledonian orogeny.

1. Introduction

The thermal and structural history of the British Caledonides is revealed by a number of lines of enquiry. A vast literature exists on the utilization of metamorphic mineral assemblages to determine the spatial and temporal distribution of pressure and temperature conditions. The relationship of key minerals to structural fabrics provides information on the relative timing of structural and thermal events [1]. However, temporal relations of growth of metamorphic minerals are difficult to establish on an absolute time scale and may be obscured by retrogressive metamorphism. Metamorphic assemblages may be diachronous and isograds marked by the first appearance of key minerals may bot be isochronous on a regional scale [2]. This limits the precision of interpretation of mineral isograds.

Another line of inquiry is the determination of times of blocking of various isotopic systems to delineate ages of intrusion, deformation and uplift. If the diffusion of daughter products of a particular system is blocked at a constant temperature, the isotopic age is the time of cooling through this temperature. "Chrontour" lines joining localities of rocks with equal apparent ages in the same isotopic system (for example K-Ar ages of muscovites) would represent the intersection of isothermal surfaces with present topography [3]. However, ages of isotopic blocking are difficult to relate to specific structural events, and partial resetting of a system or absorption of daughter products from older country rock may obscure this record.

A technique that remains to be fully explored is the application of palaeomagnetism to delineate

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aspects of the thermal and structural history of the British Caledonides. The process of thermal magnetic blocking in an assemblage of single-domain grains is analogous to the blocking of isotopic systems in minerals [4]. Thermal events may impart a magnetic overprint just as isotopic clocks may be reset. It is well documented and that the slow uplift and cooling of Precambrian mobile belts may record entire tracks of apparent polar wander paths [5–7]. Theoretical relationships for the dependence of blocking temperature and intensity of magnetization on cooling rate may allow determination of rate of cooling in certain circumstances [8]. The vector nature of palaeomagnetic data allows a direct comparison to the attitude of structural fabrics. From the reference polar wander path [9] the timing of magnetization may be determined relative to structural events. In ideal circumstances, the age of deformation may be bracketed e.g. in the Keweenawan rocks of the Lake Superior region [10].

Many potential problems may be anticipated. Magnetization may be imparted by chemical or mineralogical alternation rather than by purely thermal effects. In the absence of a comprehensive theory of chemical remanent magnetization, it is impossible to deduce uniquely the physical conditions in which the rock was magnetized from the properties of its remanance. Also, many rocks, metamorphic or otherwise, simply do not have the suitable magnetic mineralogy and grain size to record and preserve a fossil magnetization by any mechanism. The identification of metamorphic lithologies with the suitable physical properties to record the palaeomagnetic field remains a problem. Even if the palaeomagnetic field is faithfully recorded, complex polyphase deformation may render this record uninterpretable. These factors will always limit the amount and quality of information that can be derived. Nevertheless, as demonstrated below, the palaeomagnetic record of metamorphic rocks may offer insight into thermal history that would otherwise be unobtainable.

Reported here are palaeomagnetic results from intensely deformed and metamorphosed heavy mineral seams and associated psammite in the Moine of northwest Scotland. The polyphase structural and metamorphic history of these rocks probably extends back to the Precambrian [11] and the grade of regional metamorphism varies from garnet to sillimanite [12]. The palaeomagnetic remanence is not affected by structural fabric and is remarkably uniform over a region including the Morar, Knoydart and Glenelg districts. This magnetic overprint is related to the closing stages of the Caledonian orogeny.

2. Geology of the heavy mineral seams

The heavy mineral seams are of general interest because they are found in a restricted stratigraphic interval in the Morar Division which is the lowest major unit in the Moine [13]. Typically the seams occur as discontinuous layers or lenses of specularite, ilmenite, magnetite, sphene, zircon biotite and epidote, between 1 and 10 cm in thickness, within a light grey psammite which carries dark pelitic laminae. They are often found associated with relict sedimentary structures that may be used as "way-up" indicators [14]. Because of their association with sedimentary fabrics, it is thought they represent placers concentrated by fluvial processes during the deposition of the Morar Division.

The tectonic complexity of these rocks is evident at the sampled outcrops. At most localities, interference structures indicative of multiple folding, and pinched relicts of fold highes are common.

The sampling localities of the heavy mineral bands and associated psammite are shown in Fig. 1, together with major faults. The metamorphic grade ranges from garnet in the west to sillimanite in the east with isograds striking north-south. There is some evidence for the existence of retrogressed rocks of biotite grade in the middle of the Morar district [12,15]. According to Ramsay [16] the isograd surfaces probably predate the intense third phase of folding (F_3). However, Powell [11] observes two phases of metamorphism: a regional event of garnet grade affecting the entire region and either synchronous or predating the second folding phase (F_2), and a later metamorphism reaching high grade in the east and low grade in the west either coincident with or later than F_3.

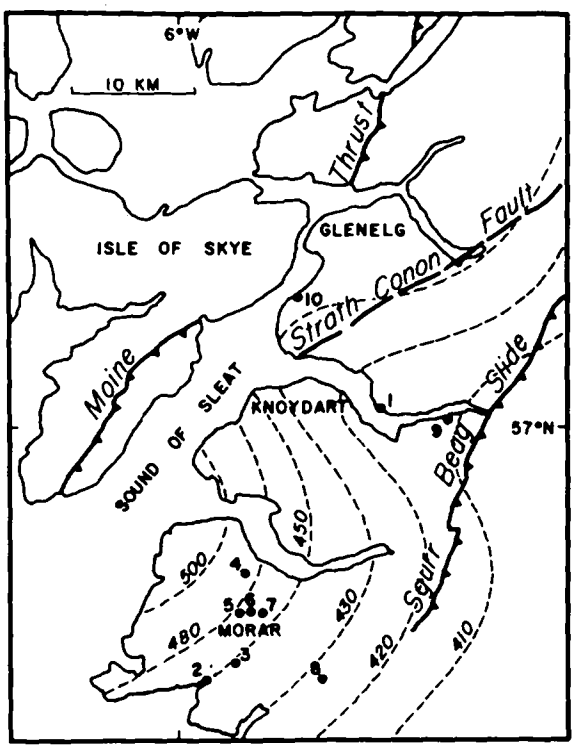


Fig. 1. Map showing sampling localities of heavy mineral seams and associated psammite with respect to major faults and muscovite and whole rock potassium-argon chrontours.

Dewey and Pankhurst [3] refer to the potassium-argon age record of this part of Scotland in the context of the regional distribution of ages in the Scottish Highlands. The distribution of muscovite and whole rock ages in the Morar, Knoydart, and Glenelg regions of Scotland is summarized by the chrontour pattern in Fig. 1 reproduced from Dewey and Pankhurst [3]. The ages are oldest on the west coast and are progressively younger toward the east. Dewey and Pankhurst [3] interpreted this pattern as reflecting differential uplift of the terrain. They speculate that the oldest ages may reflect uplift before the main phase of Caledonian metamorphism and the sinuosity of the chrontours is caused by post-cooling deformation. A principal motivation of the present study was to determine if the chrontour pattern is reflected in the magnetic record, and to determine the timing of magnetization relative to structural events.

3. Sampling and measurement

Oriented cores 2.54 cm in diameter and usually 10 cm in length were collected from the heavy mineral bands and the associated psammite using a portable coring apparatus. The sampling locations are indicated in Fig. 1. The sampling program was designed to traverse the K-Ar chrontours and to cover different grades of regional metamorphism. Orientations were derived from a sun compass when possible; otherwise magnetic bearings were taken. The locations of the bands were determined from Ramsay and Spring [14], P.W.G. Tanner (personal communication) and six inch base maps at the Institute of Geological Sciences, Edinburgh.

The cores were cut into specimens 2.3 cm long. Measurements of the magnetization were carried out with a complete results spinner magnetometer [17]. Preliminary thermal demagnetization was carried out with a mu-metal shielded thermal demagnetizer and detailed experiments were done in a large furnace in which the external field is annulled by orthogonal coils with feed-back control [18,19]. Alternating field demagnetizations were carried out using a tumbler device with a peak field of 200 mT [20].

The results of the demagnetization were plotted on orthogonal projections [21]. A detailed demagnetization sequence was carried out on each specimen to resolve the potential multicomponent nature of the NRM and to derive blocking temperatures of the various components. Components determined from linear trajectories on the orthogonal projections [22] were isolated by vector subtraction and classified according to blocking temperature and coercivity. The analysis of the vector diagrams was done by visual examination. Selected demagnetization series were analysed using a straight line-fitting algorithm [23] to determine if the selection of components was meaningful when errors of individual measurements were taken into account, and to check the visual interpretations.

4. Palaeomagnetic properties

Six of the ten localities which were sampled have very similar magnetic signatures char-

acterized by: (1) well grouped NRM's with either normal or mixed polarities; (2) maximum coercivity greater than 200 mT; and (3) multicomponent remanence as resolved by detailed thermal demagnetization.

The characteristic behaviour of the heavy mineral seams is shown in the orthogonal projection in Fig. 2a and the associated psammite in Fig. 2b. The natural remanent magnetization (NRM) is dominated by a component that is removed in a blocking temperature (T_B) interval between 590° and 630°C. A high T_B component persists up to demagnetization temperatures of 680°C having a typical intensity of 10% of NRM. The behaviour of the psammite is very similar, except that intensities of magnetization are up to two orders of magnitude less. In two localities, a third component with T_B less than 590°C is found overprinting the intermediate T_B component (Fig. 2a). A few specimens from these two localities show four components in the NRM. As exemplified by Fig. 2b, two components are present in the blocking temperature interval between NRM and 483°C. These are both represented only by two point trajectories, but the reality of these components, when considered in terms of the accuracy of the individual measurements of magnetization, is confirmed using the line fitting algorithm of Kent et al. [23].

The palaeomagnetic signatures of the other four localities are not entirely characterised by the above description. The results from locality 7 are the most exceptional. The remanence is characterised by rather low T_B (590°C maximum, see Fig. 2c) and low coercivity (5–40 mT). The mean direction of this magnetization is unique to this locality and is not comparable to the low-T_B component discussed previously because of the low coercivity. A single sample from this site shows a low-T_B component with similar physical properties but different direction overprinting a component that persists up to 667°C (Fig. 2d). This high-T_B component has a direction consistent with that in the characteristic sites.

The behaviour of the magnetization at locality 8 is very similar to that shown in Fig. 2d. The high-T_B and intermediate-T_B components are overprinted by a low-T_B/low-coercivity component

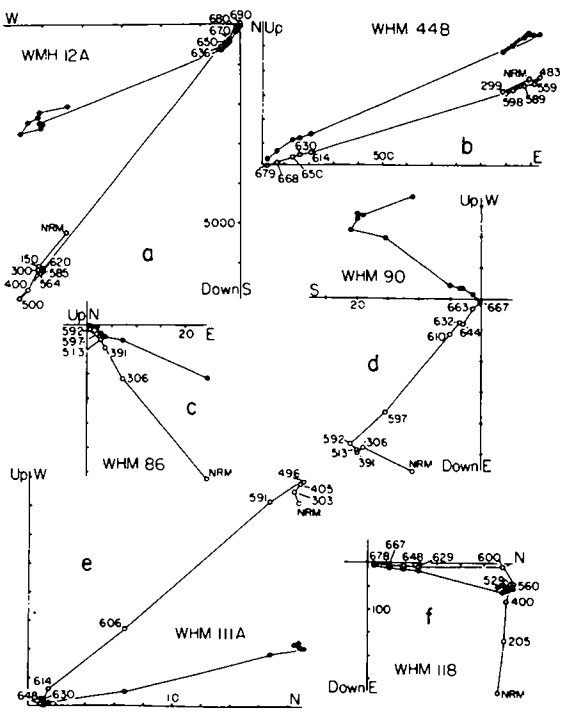


Fig. 2. Orthogonal projections [21] illustrating various aspects of the behaviour of the magnetization during thermal demagnetization, as discussed in the text. The units of the axes are 10^{-6} G = 10^{-3} A/m. Open (closed) circles denote projection on the vertical (horizontal) plane.

which has randomly distributed directions in the different specimens.

At locality 9, the heavy mineral seams have all the characteristics described for the first six sites, but the psammite record is unique. Components of the intermediate-T_B class (590–630°C) are identified in the psammite (Fig. 2e), but the directions are not the same as in the heavy mineral seams. They fall on a great circle joining endpoints defined by the typical normal and reversed directions derived from the heavy mineral seems (Fig. 3). It is most likely that this is due to normal and reversed components having indistinguishable blocking temperature spectra rather than a polarity transition. Intermediate directions characteristic of transitional behaviour do not fall on great circles, and a large number of transitions pass through steep inclinations [24]. The psammite record at this locality must represent a limitation of this lithology

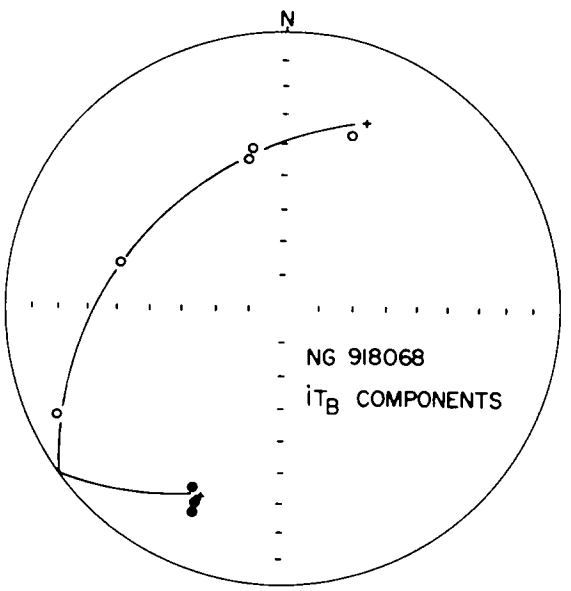


Fig. 3. Equal area projection of intermediate-T_B components isolated from locality 9. The crosses denote the mean normal and reversed directions of remanence components isolated from the heavy mineral seams. The directions of the components isolated from the psammite fall on the great circle indicated in the diagram, joining the mean normal and reversed directions from the seams. Open (closed) circles denote projection on the upper (lower) hemisphere.

as an accurate recorder rather than representing real behaviour of the geomagnetic field.

At locality 10 the host lithology is red quartzite and the seam is dull red with none of the grey specularite that characterises the other seams. This site is on the northern side of the Strath Conon Fault and may have different uplift history (D.I. Smith, personal communication). Three components of the NRM are resolved by thermal demagnetization (Fig. 2f). A low-T_B component is revealed between NRM and 560°C, an intermediate-T_B component between 600°C and 630°C, and high-T_B component that persists up to 670°C.

5. Pole positions

The mean directions, statistics and pole positions of the magnetizations derived from the heavy mineral seams and the associated psammite are summarized in Table 1 and are compared with palaeomagnetic poles derived from Great Britain and continental Europe in Fig. 4.

The palaeomagnetic signature of these rocks is best interpreted in terms of the European record. If these poles are rotated into the North American frame of reference using the parameters required for the Bullard et al. [25] restoration, they do not correspond with any Phanerozoic North American poles.

In each case, where the intermediate-T_B components are statistically distinguishable from the high-T_B components, the high-T_B pole lies to the southeast of the intermediate-T_B pole. The means of the intermediate-T_B and high-T_B components from localities 8, 9 and 10 are not statistically distinguishable and they do not exhibit this trend.

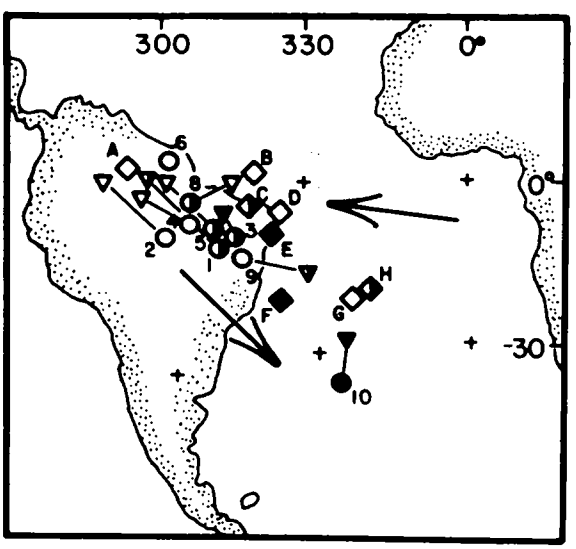


Fig. 4. Pole positions derived for the high-T_B (circles) and intermediate-T_B (triangles) components compared to European Silurian and Devonian poles. The diamonds are poles graded by Briden and Duff [9] as grade "C" or better quality: A = Welsh Old Red Sandstone, B = Lorne Plateau volcanics, C = combined Lower Old Red Sandstone lavas, D = Garabal Hill–Glen Fyne Complex, E = Arrochar Complex, F = Montmartin red beds, G = Ringerike sandstones, H = Roragen sandstones. The John O'Groats Sandstone pole is not used because it is controversial [28]. The closed symbols denote reversed polarity, the open circles normal polarity and the partially closed symbols mixed polarity. The high-T_B poles are labelled by locality and the intermediate-T_B poles are connected to the associated high-T_B pole by a solid line. The arrows denote direction of apparent polar wander.

TABLE 1

Mean directions, statistics and pole positions

Site	Class	Polarity	Remanence directions					Pole positions			
			N	$\bar{D}(°)$	$\bar{I}(°)$	k	$\alpha_{95}(°)$	lat.(°)	long.(°)	$dp(°)$	$dm(°)$
(1) NG860071	lT_B	M	15	51	−35	24	8	−1	304	5.3	9.2
	iT_B	M	17	240	36	15	10	2	300	6.8	11.6
	hT_B	M	19	220	28	15	9	−10	314	5.4	9.9
(2) NM711843	lT_B	R	9	241	21	5	26	−6	294	14.4	27.4
	iT_B	N	10	65	−29	74	6	0	293	3.6	6.6
	hT_B	N	9	50	−25	106	5	−9	304	2.9	5.4
(3) NM736856	iT_B	M	2	42	−37	32508	1	−5	315	0.7	1.2
	hT_B	M	4	39	−31	396	5	−9	317	3.1	5.6
(4) NM743932	iT_B	N	4	56	−32	297	5	−2	301	3.2	5.6
	hT_B	N	4	48	−31	27	18	−6	308	11.2	20.1
(5) NM745898	iT_B	N	4	58	−38	291	5	2	302	3.5	5.9
	hT_B	M	5	43	−34	186	6	−7	314	3.9	6.9
(6) NM749897	iT_B	N	1	51	−49	–	–	7	311	–	–
	hT_B	N	5	56	−42	103	8	4	305	6.0	9.8
(7) NM753897	lT_B	?	5	133	67	122	7	24	25	9.6	11.6
	hT_B	R	1	207	45	–	–	−3	330	–	–
(8) NM807841	iT_B	M	10	39	−44	21	11	1	319	8.6	13.8
	hT_B	M	4	48	−36	14	26	−3	309	17.5	30.2
(9) NG918068	iT_B	M	4	203	27	221	6	−16	330	3.5	6.5
	hT_B	R	2	216	26	21	57	−13	318	33.3	61.6
(10) NG778164	lT_B	N	4	219	85	91	10	50	345	19.6	19.8
	iT_B	N	4	15	−7	42	14	−28	336	7.1	14.1
	hT_B	N	4	17	9	14	25	−36	333	12.7	25.2

Site: British national grid coordinates. Class: lT_B = low blocking temperature component, iT_B = intermediate blocking temperature component, hT_B = high blocking temperature component. Polarity: M = mixed, R = reversed, N = normal. N = The number of independent results used to compute the mean; $\bar{D}$ = mean declination; $\bar{I}$ = mean inclination; k = estimate of Fisher's [29] precision parameter; α_{95} = apical half angle of the cone of 95% confidence; lat. = latitude of palaeomagnetic pole; long. = longitude of palaeomagnetic pole; dp, dm = angular lengths of the semi-axes of the oval of 95% confidence of the pole positions [30].

The high-T_B poles are comparable to poles derived for the Midland Valley volcanics (C) and Scottish Newer Granites (D and E). The intermediate-T_B poles form a streak from the high-T_B poles to the Welsh Old Red Sandstone pole (A). A younger age of acquisition of the intermediate T_B component is consistent with (though not proof of) a simple cooling model in which the terrain is slowly uplifted, passing through isothermal surfaces corresponding to the blocking temperatures of the components. The relation of the temperature of magnetization to the laboratory blocking temperature would depend on the cooling rate [8]. Alternatively, the high-T_B poles may be younger than the intermediate-T_B poles if they fall on the polar wander track going from the Welsh Old Red Sandstone pole to younger poles. The major point of interest is that the ages of magnetization are apparently much younger than is predicted from the chrontour pattern of Fig. 1. No magnetization older than late Silurian has been detected.

Pole 10 is from the dull red heavy mineral seam that occurs on the north side of the Strath Conon Fault. Its remanence is not part of the population found in the seams collected from the south side of the fault. The intermediate-T_B and high-T_B poles fall on the late Devonian segment of the British polar wander path. This may indicate very different cooling histories for the respective blocks separated by the Strath Conon Fault but with only four samples supporting this result, any conclusion as to the significance of this different signature must remain speculative.

6. Conclusions

The heavy mineral seams and associated psammite in the Moine Assemblage of the Morar, Knoydart and Glenelg districts south of the Strath Conon Fault have a characteristic palaeomagnetic signature that does not correlate with structural fabric, metamorphic grade or the pattern of potassium-argon ages. The signature is best interpreted in terms of the palaeomagnetic record from Europe and is similar to the Newer Granite suite, late Caledonian volcanics and the Old Red Sandstone. Where statistically distinguishable, the directions of the components shift systematically from high T_B to intermediate T_B implying detectable movement of the palaeomagnetic pole during the time required for the magnetization of these rocks. The low dispersion and the occurrence of normal and reversed polarities within single specimens means that the time scale of magnetization was long compared to the time scales of secular variation and reversals of the geomagnetic field.

The final interpretation of the significance of this palaeomagnetic signature depends on the mechanism of acquisition. At this point, it is not possible to assign a chemical or thermal origin to the characteristic magnetizations.

If the magnetization is purely of thermal origin, the high-T_B component must be older than the intermediate-T_B component. No matter what the cooling rate [8] or the time of isothermal heating [26], laboratory blocking temperatures in haematite of 600°C or more imply ambient rock temperatures of 600°C or more. Estimating the age of magnetization as late Silurian to early Devonian, the temperature of the rocks is inferred to have been much higher at this time than might be expected from the potassium-argon data. The chrontours of Dewey and Pankhurst [3] imply temperatures of 300°C at times ranging from 500 to 410 m.y., whereas a thermal origin of the NRM would imply temperatures of 600°C as late as 410–390 m.y.

However, if the magnetization is of chemical origin, the temperature of the rock at the time of acquisition could be much lower than the laboratory blocking temperature. In this case, the high-T_B component would not necessarily be older than the intermediate-T_B component. An older age for the intermediate-T_B components would in fact fit Turnell's [27] interpretation of the polar wander path which shows the return path going slightly south of poles B through E in Fig. 4.

Either mode of origin of NRM requires a regional mechanism to remagnetize the Moine on the scale of Fig. 1. One possible mechanism could be Newer Granite Activity. A.J. Highton (personal communication) suggests that vein complexes and associated minor intrusives in northwest Scotland may represent the highest (or lowest) levels of granitic bodies. A large newer granite mass be-

neath (or above) the present erosion surface could have reset the palaeomagnetic record of the heavy mineral seams in the observed fashion. If this is indeed the case, why is the existence of such a body not reflected in the isotopic record?

It is difficult to reconcile the potassium-argon record with any remagnetization mechanism that would raise the temperature of the Moine near the blocking temperature of argon in micas and whole rocks. It is possible all of the magnetizations are low-temperature chemical remanent magnetizations of Siluro-Devonian age obscuring all pre-existing remanences. But the palaeomagnetic overprint is surprisingly consistent on a regional scale and is found in two very different rocks, i.e., the heavy mineral seams and the host psammite. It remains puzzling that the remagnetization event is not reflected in the potassium-argon age pattern.

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